
Assignment 02

1. The Bubble Sort algorithm has a time complexity of $O(n^2)$ in its worst and average cases, which makes it inefficient for large datasets. How we can optimise the Bubble Sort algorithm
And implement the code of this optimised bubble sort algorithm

2.You are given an **ArrayList** containing a sequence of elements. try to reverse the order of elements in the **ArrayList** in-place(in the same arrayList) without using the built-in **Reverse**. Implement a function that takes the **ArrayList** as input and modifies it to have the reversed order of elements.

3. Given an array consists of numbers with size N and number of queries, in each query you will be given an integer X, and you should print how many numbers in array that is greater than X.

Ex:

Input

3 3

//Size of array , number of queries

11 5 3

//Array

1

//Query1

5

//Query2

13

//Query 3

Output

3

//11,5,3

1

//11

0

4. Given a number N and an array of N numbers. Determine if it's palindrome or not.

Ex:

Input:

5

1 3 2 3 1

Output:

YES

5. Given a Queue, implement a function to reverse the elements of a queue using a stack.

6. Given a Stack, implement a function to check if a string of parentheses is balanced using a stack.

Ex:

Input:

[(){}]

Output:

Balanced

7. Given an array, implement a function to remove duplicate elements from an array.

8. Given an array list , implement a function to remove all odd numbers from it.

9. Implement a queue that can hold different data types.

And insert the following data:

```
queue.Enqueue(1)
queue.Enqueue("Apple")
queue.Enqueue(5.28)
```

10. Create a function that pushes a series of integers onto a stack. Then, search for a target integer in the stack. If the target is found, print a message indicating that the target was found how many elements were checked before finding the target ("Target was found successfully and the count = 5"). If the target is not found, print a message indicating that the target was not found("Target was not found").

Note : take the target as input from the user

11. Given two arrays, find their intersection. Each element in the result should appear as many times as it shows in both arrays.

Ex :

Input :

5 , 3

[1,2,3,4,4] , [10,4,4]

Output :

[4,4]

12. Given an ArrayList of integers and a target sum, find if there is a contiguous sub list that sums up to the target.

Ex :

Input :

[1, 2, 3, 7, 5]

12

Output :

[2, 3, 7]

13. Given a queue reverse first K elements of a queue, keeping the remaining elements in the same order

Ex :

Input :

[1 , 2 , 3 , 4 ,5]

K = 3

Output :

[3 , 2 , 1 , 4 ,5]